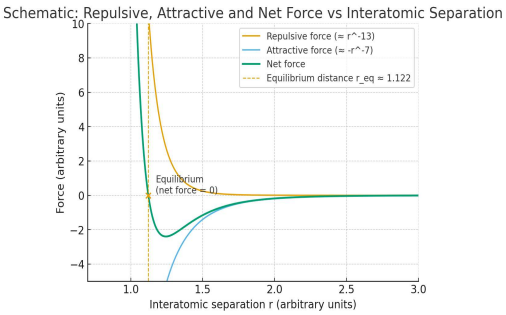
QP CODE: 25900562 Scheme and Answer Key for Valuation

MGU-UGP(Honours)Regular Examination, October 2025
MG3DSEPHY204- INTRODUCTION TO MATERIAL SCIENCE
Time: -1hour 30-minute **Maximum Marks: - 50**

Part A
Answer any 10 questions. Each question carries 2 marks

Question	Marks
1. *What do you mean by smart materials? Which are the two components of smart materials?* Materials that can sense and respond to external stimuli (stress, temperature, magnetic or electric field). Components: sensor and actuator.	Definition-1mark Two components-1 marks One component 0.5mark
2. *Why are amorphous solids considered isotropic in nature?* Atoms are arranged randomly; physical properties are the same in all directions.	Any two points 2marks
3. *Find the Miller indices of a plane that intercepts the axes at (a, a, a).* Intercepts along x, y, z at 1a → reciprocals (1,1,1) → Miller indices (111).	All steps with final answer-2 marks Partial attempt -1mark
4. *Schematically plot attractive, repulsive and net force versus interatomic separation for two atoms or ions. Also, mark the equilibrium distance.*  The plot shows attractive and repulsive forces vs distance; the net force curve	Correct plot with markings 2marks Plot only 1 mark Plot with partial markings 1.5 marks

intersects zero at equilibrium separation.	
5. *What is solvus line?* Line separating single solid solution and two-phase regions in a phase diagram; defines solubility limit.	Correct definition 2marks Diagrammatic representation -2 marks
6. *What is dendritic structure?* Tree-like crystal growth pattern during solidification due to non-uniform cooling.	Correct definition 2marks
7. *Differentiate between homogeneous and heterogeneous nucleation.* Homogeneous: within bulk; no impurities. Heterogeneous: on surfaces or impurities; lower energy barrier.	Homogeneous nucleation 1 mark Heterogeneous nucleation 1 mark
8. *What is the effect of higher activation energy on transformation kinetics?* Higher activation energy → slower transformation rate.	Correct answer 2marks
9. *What is the critical radius of a nucleus, and what does it signify?* Minimum radius of stable nucleus; below it, nuclei dissolve; above it, growth occurs.	Critical radius 1mark Significance 1 mark
10. *What role does undercooling or supercooling play in controlling phase transformation kinetics?* Greater undercooling increases nucleation rate and refines grain size.	Correct answer 2 marks
11. *Name the test used to measure notch toughness.* Charpy or Izod impact test.	2marks
12. *Explain Hooke's law.* Stress is directly proportional to strain within the elastic limit ($\sigma = E\varepsilon$).	Correct definition with expression 2marks Definition or expression only 1 mark
13. *Define brittle fracture.* Fracture with little or no plastic deformation; sudden failure.	2 marks

14. *Define corrosion and give one common example*. Deterioration of a metal by chemical reaction with environment, e.g., rusting of iron.	Definition 1 mark Example 1 mark

Part B

Answer any 6 questions. Each question carries 5 marks [5 x 6 =30]

Question	Marks
15. Compare ionic and covalent bonding with respect to bond energy, melting point, and conductivity. Ionic: high bond energy, high melting point, non-conductive as solids; Covalent: moderate bond energy, variable melting point, poor conductors.	Correct answer for all three 5 marks Deduct marks for partial answers
16. Explain the seven crystal systems with their axial relationships. Cubic (a=b=c, α=β=γ=90°); Tetragonal (a=b≠c, α=β=γ=90°); Orthorhombic (a≠b≠c, α=β=γ=90°); Hexagonal (a=b≠c, α=β=90°, γ=120°); Trigonal (a=b=c, α=β=γ≠90°); Monoclinic (a≠b≠c, α=γ=90°, β≠90°); Triclinic (a≠b≠c, α≠β≠γ≠90°).	All seven crystal systems 5 marks 0.75 marks for each correct system
17. Discuss how impurities influence diffusion processes in solids. Support your answer with relevant examples. Impurities can accelerate or retard diffusion; substitutional and interstitial mechanisms, e.g., C in Fe increases hardness but slows atomic motion.	Effect of impurities-3 marks Substantiation with example 2 marks Example only 1 mark
18. What is the difference between the states of phase equilibrium and metastability? Equilibrium: lowest free energy state; Metastable: temporary non-equilibrium, but appears stable until disturbed.	Correct answers for both 5 marks Any one 2.5 mark
19. Apply the Gibbs Phase Rule to a binary phase diagram and discuss how it helps in determining the number of degrees of freedom in different regions of the diagram. Phase rule: F = C – P + 2; For binary (C=2): F = 4 – P; two-phase F=2; helps	Gibbs phase rule 2 marks Determination of degrees of freedom-2

predict variable control.	marks Various regions with different variables 1 mark
20. *Explain how compressive residual stresses are beneficial for fatigue resistance.* Compressive stresses reduce surface crack initiation and growth, enhancing fatigue life (e.g., shot peening).	Correct answer -5 marks
21. *With examples, give the applications of materials with high ductility and low ductility.* High ductility: copper (wires); Low ductility: cast iron (brittle components).	Example 1 mark, application with explanation 1.5 marks for each
22. *Explain the mechanism of oxidation in metals and derive the parabolic rate law for oxidation kinetics. Discuss its significance in understanding corrosion resistance.* Oxidation forms protective oxide film; rate $\propto 1/t$; parabolic law ($x^2 = k_p t$); low rate indicates high resistance	Oxidation mechanism 1 mark Parabolic oxidation law-1 mark Derivation 2 marks Significance 1 mark